

and tested on datasets curated for a particular subject matter.

In *dataset-agnostic* scenarios, where the model is trained across multiple datasets, we observed two instances with higher Accuracy and lower FAR/FRR when training incorporated a mix of keystroke sequences from the SBU dataset (S) and either the Buffalo (B) or Proposed (P) datasets. This approach maximizes the training data pool, resulting in improved model performance. The model performs best when trained on a combination of the SBU and Buffalo datasets and tested on the Proposed dataset, achieving an accuracy of 73.23%, an F_1 score of 72.65%, FAR of 27.95%, and FRR of 40.22%. However, in cases where the training data is less diverse, both the accuracy and F_1 score fall below 60%, with FAR and FRR exceeding 40% and 45%, respectively, as illustrated in Table 3. These findings suggest that TypeNet requires more comprehensive data to generalize effectively across different environments. However, with a larger dataset, the model’s performance significantly improves, highlighting the importance of extensive training data to enhance its robustness.

4.1. Discussion

The proposed TypeNet-based plagiarism detector exhibited varied error rates across different data collection scenarios and contexts. While achieving promising performances in scenario-specific settings, the detector struggled to generalize across scenarios. This aligns with findings from previous keystroke studies, including those involving TypeNet [49, 50]. Further research is needed to develop a model generalizing across various datasets and collection environments.

A recent study indicates that simple statistical models outperform advanced deep learning models like TypeNet in keystroke classification [51]. Consequently, we plan to explore the effectiveness of latency and word-level features and classifiers such as Instance-based Tail Area Density (ITAD), SVM, kNN, and Random Forest [33, 52]. Additionally, we intend to incorporate strategies used in recent AI vs. Human detectors [7–13], alongside content-independent linguistic features (such as intensifiers, hedging, politeness markers, negations, filler words, function words), and leverage transformer-based architectures [53] in future developments. Access to an individual’s keystroke data beforehand will allow us to create personalized models to enhance the system’s accuracy and adaptability.

While we have reported on the performance of the TypeNet architecture, a recent study [51] prompts us to evaluate traditional keystroke modeling approaches in future research. We simplified our experimental setup by assuming copy-paste actions would be disabled using tools such as lockdown browsers during examinations. However, this assumption may not hold in all examination settings, as some may allow copy-pasting within the test editor. Monitoring these actions and clipboard data could enhance our method’s performance.

Moreover, a detailed theoretical analysis of the distinguishability of typing patterns would provide further insights to our proposed method. Nevertheless, existing research [48] indicates that bona fide (free) writing typically involves shorter bursts of typing, longer pauses, and more frequent revisions compared to assisted (transcription-like) tasks. These observations underscore the necessity for further investigation

into typing pattern variability.

5. Limitations

While this study underscores the potential of keystroke dynamics for detecting academic dishonesty within the context of large language models (LLMs), several limitations necessitate further investigation. Firstly, the findings presented are based on a limited number of users and datasets created in controlled environments. These datasets may not adequately represent the diverse typing behaviors of a broader population, thereby impacting the model’s accuracy and applicability across various demographic groups and educational settings, such as creative writing, timed responses, or coding tasks. Moreover, a high rate of false positives could lead to wrongful accusations and subsequent distrust in the detection system, highlighting the need for more sophisticated models to reduce errors.

Additionally, the method’s effectiveness varied significantly with external factors such as keyboard layout, academic contexts, and individual typing habits. Future research should focus on developing more robust models that can adapt to these variables, ensuring consistent performance across different hardware configurations and user backgrounds.

6. Conclusions and future work

We explored the potential of keystroke dynamics to detect instances of academic dishonesty facilitated by advanced generative AI tools. Based on keystroke dynamics, our model effectively distinguished between bona fide and AI-assisted writings at the keystroke level, showing promising performance across various testing scenarios. These results underscore the value of keystroke dynamics as a reliable supplement to traditional plagiarism detection tools, particularly crucial in an era dominated by large language models (LLMs) capable of generating human-like texts.

The study also highlights the operational challenges of implementing keystroke dynamics across diverse contexts. Although the model demonstrated reasonable effectiveness under controlled experimental conditions, applying it in real-world settings necessitates further extensive research. Moving forward, we aim to broaden our research to encompass a wider array of users, educational settings, and forms of academic dishonesty, including AI assistance, integration of internet content, and copy-paste-edit behaviors. Additionally, we plan to refine our detection algorithms to improve their accuracy and effectiveness, ensuring they are adaptable across a diverse range of user demographics and academic contexts.

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References

- [1] KR Rao. Plagiarism, a scourge. *Current Science*, 2008.
- [2] Irene Anderson. Avoiding plagiarism in academic writing. *Nursing standard*, 2009.
- [3] Diane Pecorari. Academic writing and plagiarism. *Bloomsbury Publishing*, 2008.
- [4] Miguel Roig. Avoiding plagiarism, self-plagiarism, and other questionable writing practices: A guide to ethical writing. *The Office of Research Integrity (ORI)*, 2015.
- [5] Muntasir Hoq, Yang Shi, Juho Leinonen, Damilola Babalola, Collin Lynch, Thomas Price, and Bitu Akram. Detecting chatgpt-generated code submissions in a cs1 course using machine learning models. *SIGCSE 2024*, 2024.
- [6] Haijun Zhang and Tommy WS Chow. A coarse-to-fine framework to efficiently thwart plagiarism. *Pattern Recognition*, 2011.
- [7] Dongqi Pu and Vera Demberg. Chatgpt vs. human-authored text: Insights into controllable text summarization and sentence style transfer. *arXiv preprint arXiv:2306.07799*, 2023.
- [8] Liam Dugan, Daphne Ippolito, Arun Kirubakaran, Sherry Shi, and Chris Callison-Burch. Real or fake text?: Investigating human ability to detect boundaries between human-written and machine-generated text. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2023.
- [9] Steffen Herbold, Annette Hautli-Janisz, Ute Heuer, Zlata Kikteva, and Alexander Trautsch. A large-scale comparison of human-written versus chatgpt-generated essays. *Scientific Reports*, 2023.
- [10] J. Elliott Casal and Matt Kessler. Can linguists distinguish between chatgpt/ai and human writing?: A study of research ethics and academic publishing. *Research Methods in Applied Linguistics*, 2023.
- [11] Tony Berber Sardinha. Ai-generated vs human-authored texts: A multidimensional comparison. *Applied Corpus Linguistics*, 2024.
- [12] Hosam Alamleh, Ali Abdullah S. AlQahtani, and AbdElRahman ElSaid. Distinguishing human-written and chatgpt-generated text using machine learning. In *2023 Systems and Information Engineering Design Symposium (SIEDS)*, 2023.
- [13] Niful Islam, Debopom Sutradhar, Humaira Noor, Jarin Tasnim Raya, Monowara Tabassum Maisha, and Dewan Md Farid. Distinguishing Human Generated Text From ChatGPT Generated Text Using Machine Learning. *arXiv e-prints*, 2023.
- [14] Varun Khurana, Yaman Kumar, Nora Hollenstein, Rajesh Kumar, and Balaji Krishnamurthy. Synthesizing human gaze feedback for improved NLP performance. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, 2023.
- [15] Paul Prasse, David Reich, Silvia Makowski, Tobias Scheffer, and Lena Jäger. Improving cognitive-state analysis from eye gaze with synthetic eye-movement data. *Computers and Graphics*, 2024.
- [16] Fabian Monroe and Aviel D. Rubin. Keystroke dynamics as a biometric for authentication. *Future Generation Computer Systems*, 2000.
- [17] Rick Joyce and Gopal Gupta. Identity authentication based on keystroke latencies. *Commun. ACM*, 1990.
- [18] S. Bleha, C. Slivinsky, and B. Hussien. Computer-access security systems using keystroke dynamics. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 1990.
- [19] S Press. Authentication by keystroke timing: Some preliminary results. *Rand Report*, 1980.
- [20] Pin Shen Teh, Andrew Beng Jin Teoh, Shigang Yue, et al. A survey of keystroke dynamics biometrics. *The Scientific World Journal*, 2013.
- [21] Rajesh Kumar, Vir V. Phoha, and Abdul Serwadda. Continuous authentication of smartphone users by fusing typing, swiping, and phone movement patterns. In *2016 IEEE 8th International Conference on Biometrics Theory, Applications, and Systems (BTAS)*, 2016.
- [22] Salha M Alzahrani, Naomie Salim, and Ajith Abraham. Understanding plagiarism linguistic patterns, textual features, and detection methods. *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)*, 2011.
- [23] Paul Deane and Mo Zhang. Exploring the feasibility of using writing process features to assess text production skills. *ETS Research Report Series*, 2015.
- [24] Nancy Agarwal, Nils Folvik Danielsen, Per Kristian Gravdal, and Patrick Bours. Contract cheat detection using biometric keystroke dynamics. In *2022 20th International Conference on Emerging eLearning Technologies and Applications (ICETA)*, 2022.
- [25] Elena A Kochegurova and Roman P Zateev. Hidden monitoring based on keystroke dynamics in online examination system. *Programming and Computer Software*, 2022.
- [26] Peter Kimani Mungai and Runhe Huang. Using keystroke dynamics in a multi-level architecture to protect online examinations from impersonation. In *2017 IEEE 2nd International Conference on Big Data Analysis (ICBDA)*. IEEE, 2017.
- [27] John R Hayes. Modeling and remodeling writing. *Written communication*, 2012.
- [28] Linda Flower and John R Hayes. A cognitive process theory of writing. *College composition and communication*, 1981.
- [29] Kiran S Balagani and NEW YORK INST OF TECH OLD WESTBURY. Investigating cognitive rhythms as a new modality for continuous authentication. Technical report, Tech. Rep. DTIC Document, 2013.
- [30] Mo Zhang, Hongwen Guo, and Xiang Liu. Using keystroke analytics to understand cognitive processes during writing. *International Educational Data Mining Society*, 2021.

- [31] Yee Mei Lim, Aladdin Ayeshe, and Martin Stacey. Using mouse and keyboard dynamics to detect cognitive stress during mental arithmetic. In *Intelligent Systems in Science and Information 2014: Extended and Selected Results from the Science and Information Conference 2014*, 2015.
- [32] Ahmed Patel, Kaveh Bakhtiyari, and Mona Taghavi. Evaluation of cheating detection methods in academic writings. *Library Hi Tech*, 2011.
- [33] Firuz Kamalov, Hana Sulieman, and David Santandreu Calonge. Machine learning-based approach to exam cheating detection. *Plos one*, 2021.
- [34] M Roa'a, Ibtisam A Aljazeera, and AHM Alaidi. Automated cheating detection based on video surveillance in the examination classes. *International Journal of Interactive Mobile Technologies*, 2022.
- [35] Leslie Ching Ow Tiong and HeeJeong Jasmine Lee. E-cheating prevention measures: detection of cheating at online examinations using deep learning approach—a case study. *arXiv preprint arXiv:2101.09841*, 2021.
- [36] Blaine Ayotte, Mahesh Banavar, Daqing Hou, and Stephanie Schuckers. Fast free-text authentication via instance-based keystroke dynamics. *IEEE Transactions on Biometrics, Behavior, and Identity Science*, 2020.
- [37] Fabian Monrose, Michael K Reiter, and Susanne Wetzel. Password hardening based on keystroke dynamics. In *Proceedings of the 6th ACM Conference on Computer and Communications Security*, 1999.
- [38] Vishaal Udandara, Mohit Agrawal, Rajesh Kumar, and Rajiv Ratn Shah. On the inference of soft biometrics from typing patterns collected in a multi-device environment. In *2020 IEEE Sixth International Conference on Multimedia Big Data (BigMM)*, 2020.
- [39] Mirella Monaro, Claudia Galante, Riccardo Spolaor, et al. Covert lie detection using keyboard dynamics. *Scientific Reports*, 2018.
- [40] Ritwik Banerjee, Song Feng, Jun Seok Kang, and Yejin Choi. Keystroke patterns as prosody in digital writings: A case study with deceptive reviews and essays. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Doha, Qatar, 2014. Association for Computational Linguistics.
- [41] Aythami Morales, Alejandro Acien, Julian Fierrez, John V. Monaco, Ruben Tolosana, Ruben Vera, and Javier Ortega-Garcia. Keystroke biometrics in response to fake news propagation in a global pandemic. In *2020 IEEE 44th Annual Computers, Software, and Applications Conference (COMPSAC)*, 2020.
- [42] Alvin Kuruvilla, Rojanaye Daley, and Rajesh Kumar. Spotting fake profiles in social networks via keystroke dynamics. In *2024 IEEE 21st Consumer Communications and Networking Conference (CCNC)*, 2024.
- [43] Ioannis Tsimperidis, Paul D Yoo, Kamal Taha, Alexios Mylonas, and Vasilis Katos. R 2 bn: An adaptive model for keystroke-dynamics-based educational level classification. *IEEE Transactions on Cybernetics*, 2018.
- [44] Barbara Plank. Predicting authorship and author traits from keystroke dynamics. In *Proceedings of the Second Workshop on Computational Modeling of People's Opinions, Personality, and Emotions in Social Media*, 2018.
- [45] Robert Goodman, Matthew Hahn, Madhuri Marella, Christina Ojar, and Sandy Westcott. The use of stylometry for email author identification: A feasibility study. *Proc. Student/Faculty Research Day*, 2007.
- [46] Yan Sun, Hayreddin Ceker, and Shambhu Upadhyaya. Shared keystroke dataset for continuous authentication. In *2016 IEEE International Workshop on Information Forensics and Security (WIFS)*, 2016.
- [47] Heather Lettner-Rust. Questions of transfer: Writers' perspective on familiar/unfamiliar writing tasks in a capstone writing course. *Psychology*, 2010.
- [48] Kelly Trezise, Tracie Ryan, Paula de Barba, and Gregor Kennedy. Detecting academic misconduct using learning analytics. *Journal of Learning Analytics*, 2019.
- [49] Alejandro Acien, Aythami Morales, John V. Monaco, Ruben Vera-Rodriguez, and Julian Fierrez. Typenet: Deep learning keystroke biometrics. *IEEE Transactions on Biometrics, Behavior, and Identity Science*, 2022.
- [50] Patrick Bours and Jørgen Ellingsen. Cross keyboard keystroke dynamics. In *2018 1st International Conference on Computer Applications and Information Security (ICCAIS)*. IEEE, 2018.
- [51] Ahmed Wahab and Daqing Hou. When simple statistical algorithms outperform deep learning: A case of keystroke dynamics [when simple statistical algorithms outperform deep learning: A case of keystroke dynamics]. In *Proceedings of the 12th International Conference on Pattern Recognition Applications and Methods ICPRAM*, 2023.
- [52] Blaine Ayotte, Mahesh Banavar, Daqing Hou, and Stephanie Schuckers. Fast free-text authentication via instance-based keystroke dynamics. *IEEE Transactions on Biometrics, Behavior, and Identity Science*, 2020.
- [53] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 2017.